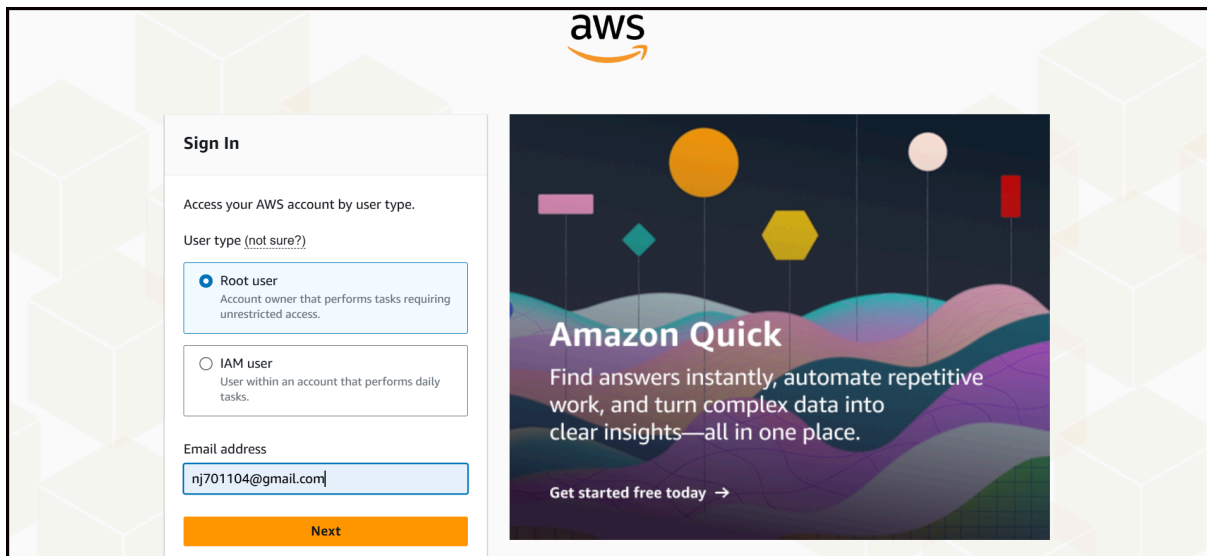


AWS - IAM Practical

Here are the **clear steps to create an IAM user with console-only access** in Amazon Web Services (AWS):

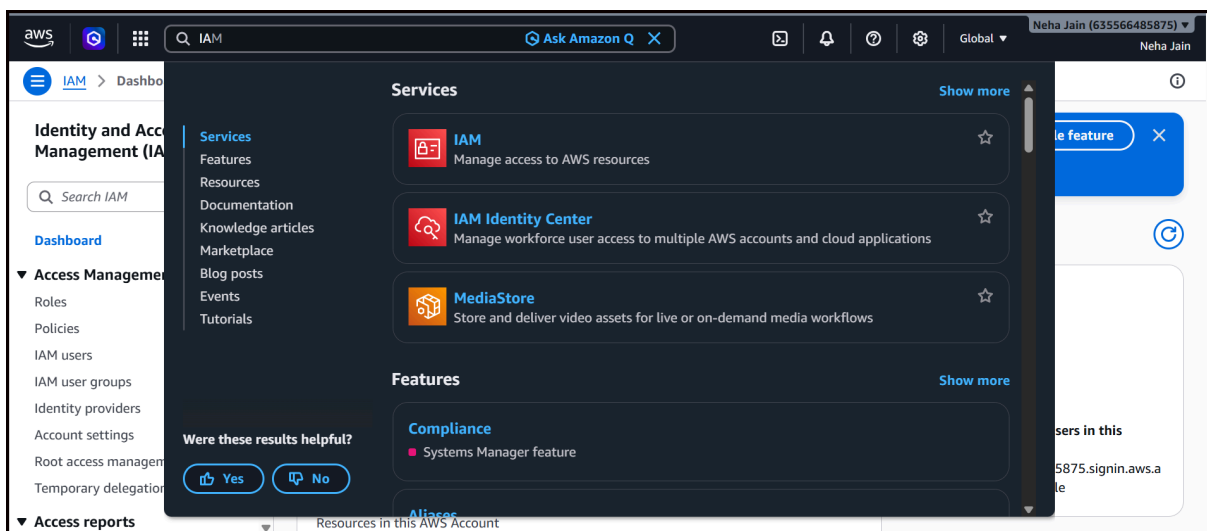
Step 1: Sign in to AWS

- Go to the AWS Management Console
- Sign in as the **root user** or an IAM admin



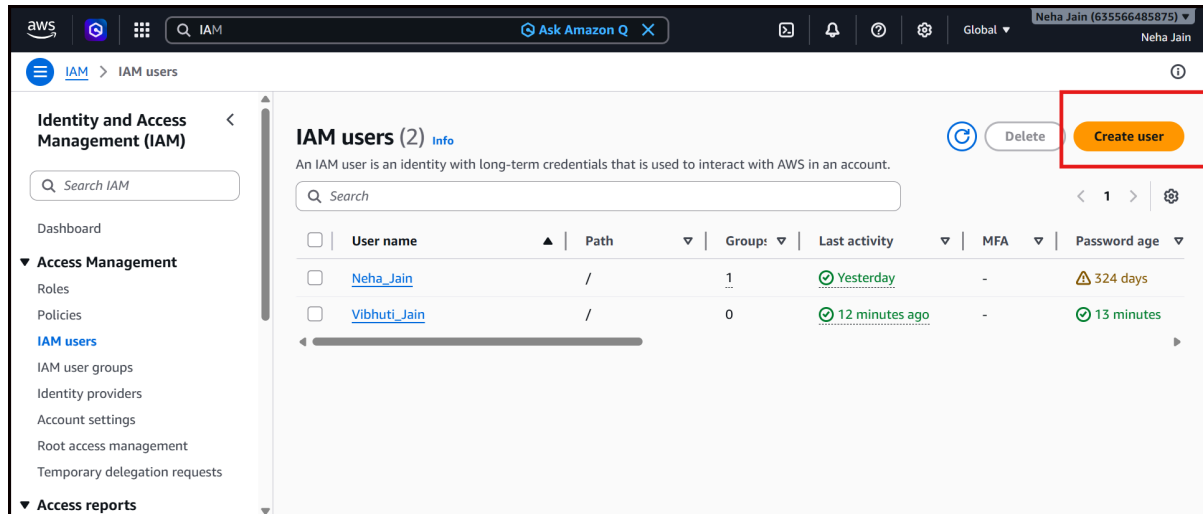
Step 2: Open IAM Service

- Search for IAM in the services bar
- Click IAM (Identity and Access Management)



Step 3: Start Creating a User

- In the left panel, click Users
- Click Create user

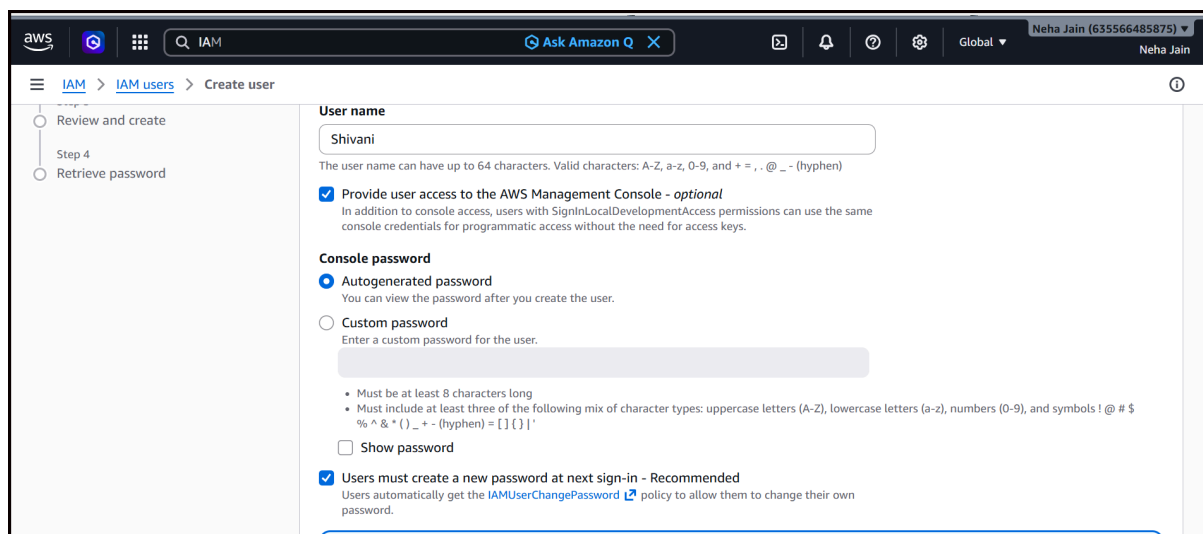


Step 4: Set User Details

- Enter a User name
- Under Select AWS access type:
 - ☒ Check: Provide user access to the AWS Management Console
- Auto-generated password OR
- Custom password

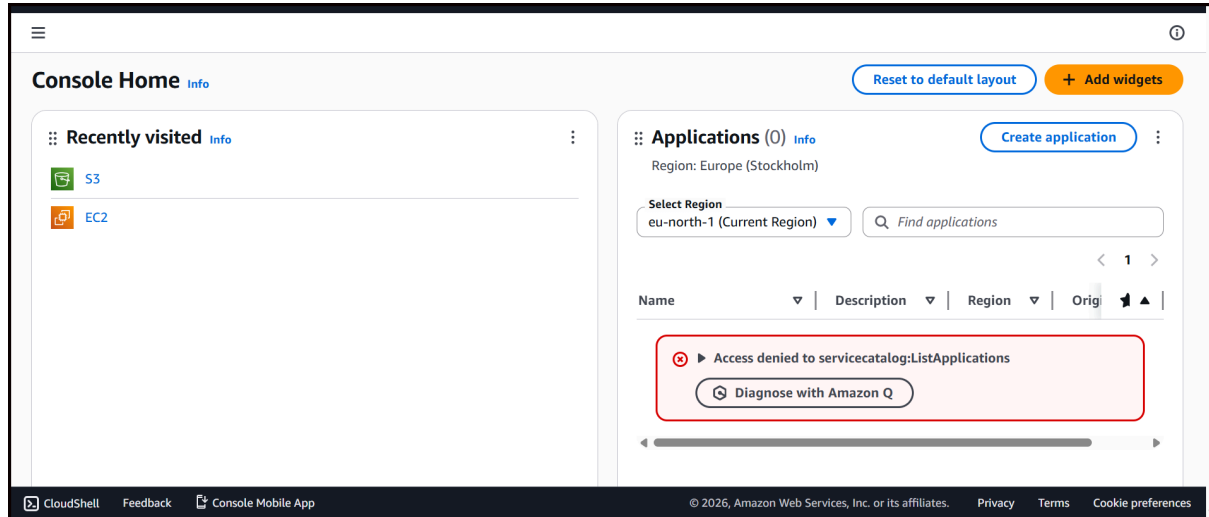
Optionally check:

- “User must create a new password at next sign-in.”

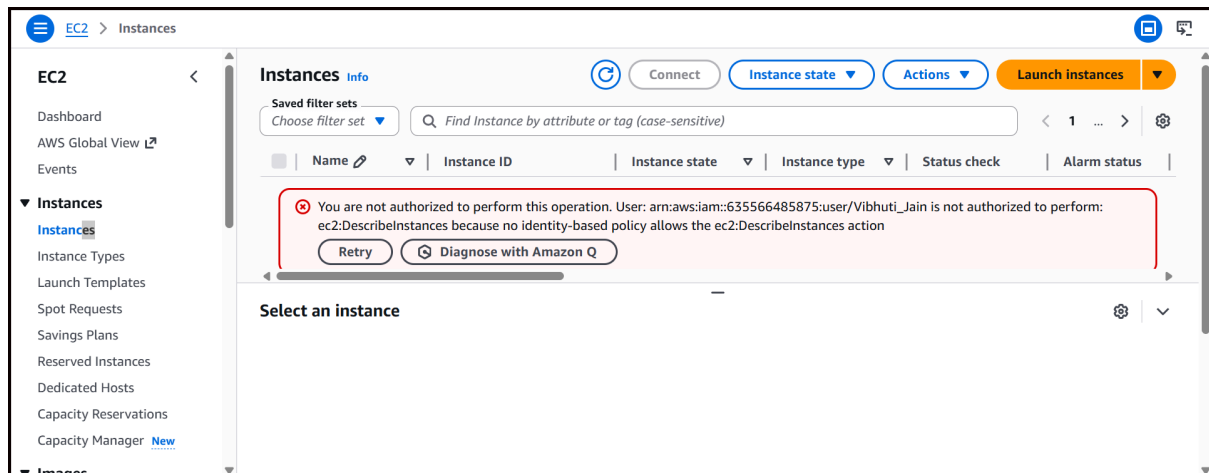


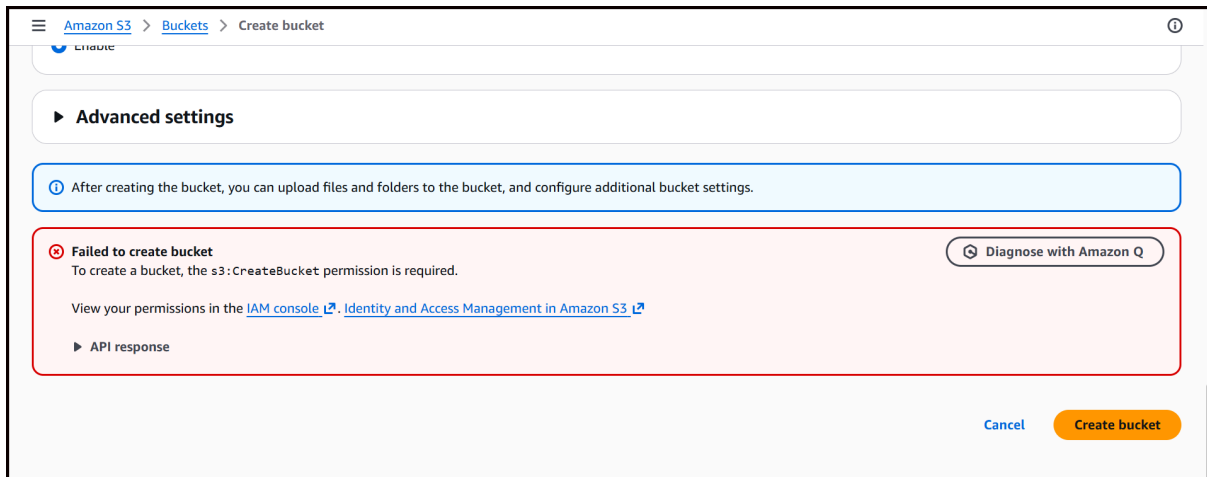
Then click Next to create a user, with no permissions for now. Our user only has access to see the AWS management console.

Now, try to log in to the console by using the credentials



See, we can access the management console. Now, let's try to access the services.

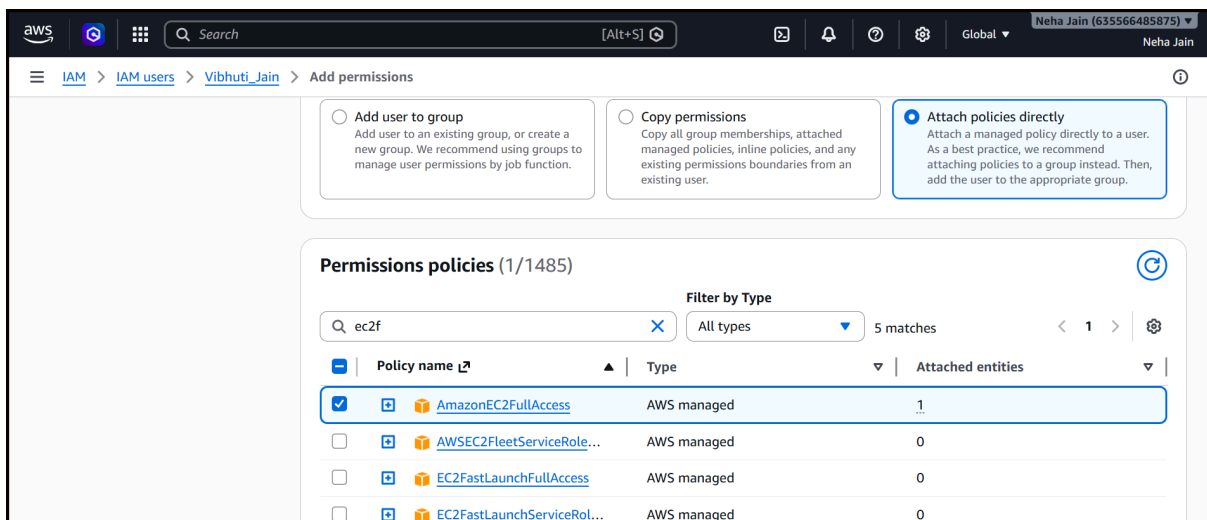
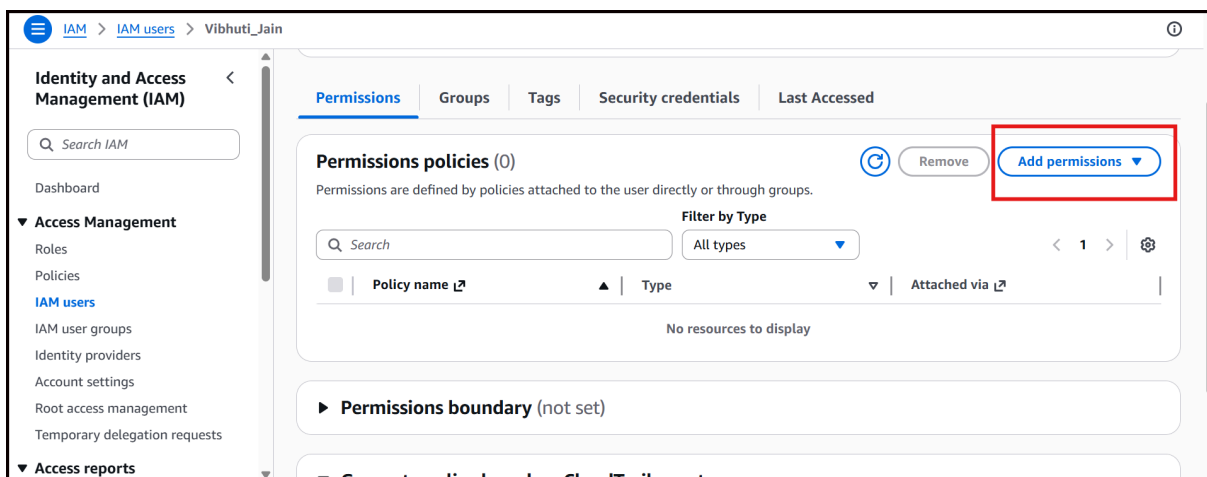




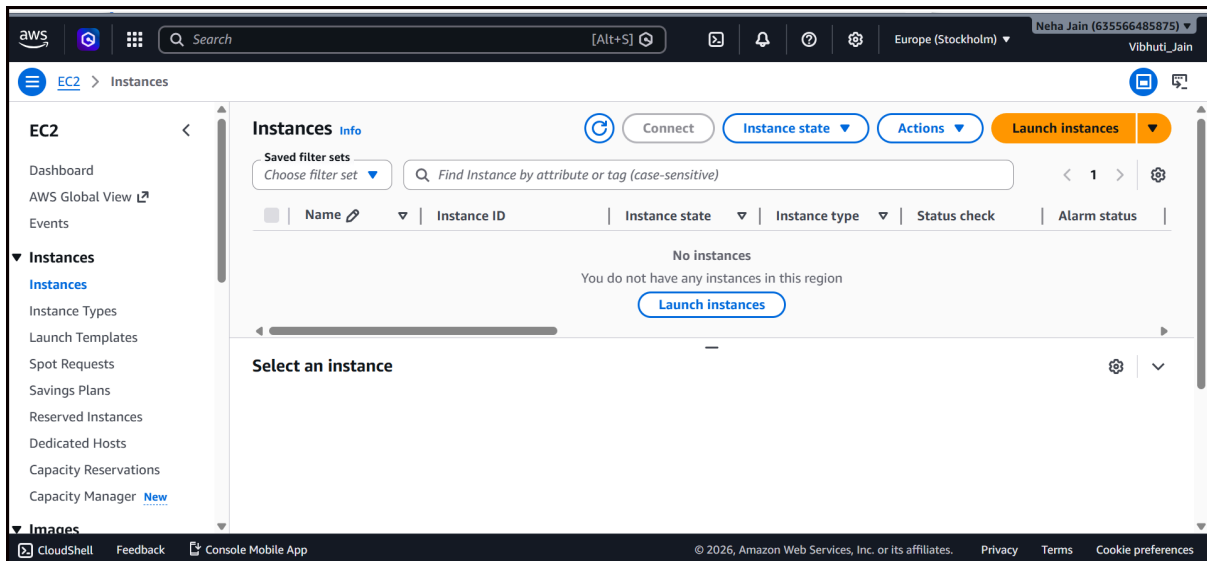
We can see that I am not able to access EC2 and S3 services in the AWS account because I didn't give any permission to the user to access the resources.

Step 5: Set Permissions: You have 3 options to set the permissions

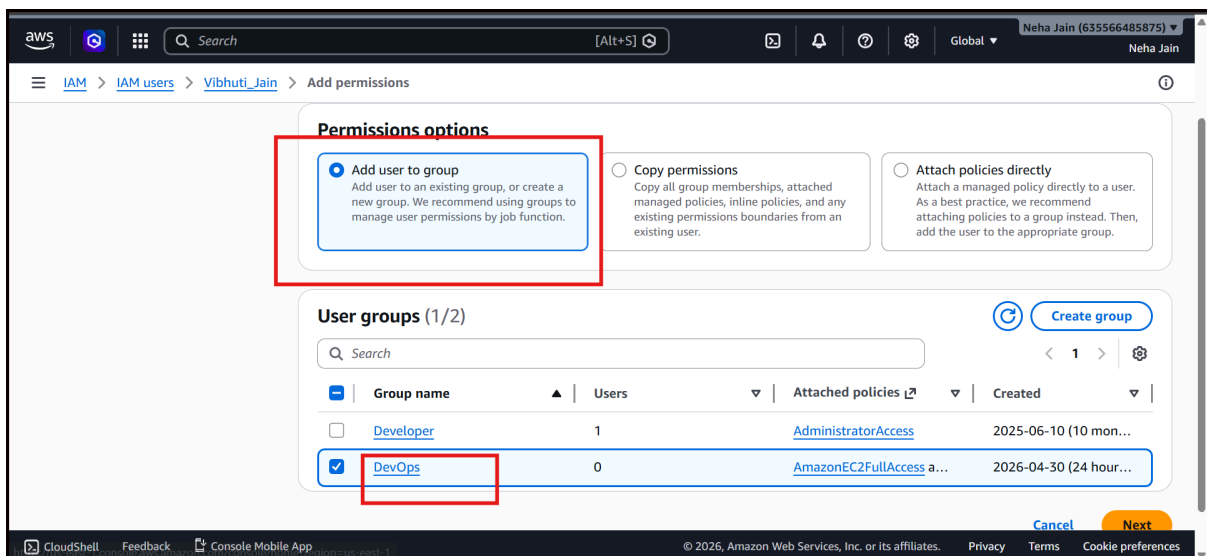
1. Add permission directly to the user - Attached Policies Directly (not recommended because we can set only up to 10 managed permissions to the users)



So, now you can see that we can access the EC2 resource because the root user granted access.



2. Add user to group (recommended):



Select an existing group (e.g., **Admins**, **Developers**)

OR create a new group and attach policies

3. **Copy permissions** from another user - Suppose you have a user “Shivani” and you want to give the exact permissions to the other user as well, you can copy the exact permissions to the other user.

❖ **User** - In Amazon Web Services (AWS), an IAM user is simply an identity you create to represent a person or application that needs access to AWS resources.

What an IAM User Includes: Each IAM user typically has:

- Username
- Password (for console login)
- Access keys (for CLI/API, if enabled)
- Permissions (what they are allowed to do)

Why IAM Users Are Used: Instead of sharing the root account, you:

- Create separate users for each person/system
- Assign only the permissions they need
- Track actions for security and auditing

Key Concept

IAM users follow the principle of:

👉 **Least Privilege** (give only necessary access)

- ❖ **Least Privilege** - Giving a user only the minimum permissions required to do their job — nothing more.

Simple Explanation

Instead of giving full access to everything, you:

- Allow only specific actions
- On specific resources
- For a specific purpose

Example

❌ Not least privilege:

- Giving a user **AdministratorAccess** (full control of AWS)

✅ Least privilege:

- Giving access only to:
 - Read files from S3
 - Or start/stop EC2 instances (if needed)

♦ Why It Matters

- 🛡️ Improves security (limits damage if account is compromised)

-  Reduces mistakes (users can't accidentally delete or modify unrelated resources)
-  Better control & auditing
-
- ❖ **Groups** - A collection of IAM users that share the same permissions.

Simple Explanation

Instead of assigning permissions to each user one by one, you:

- Create a **group**
- Attach permissions (policies) to the group
- Add users to that group

👉 All users in the group automatically get those permissions

♦ Example

You create a group called **Developers**:

- Give the group access to:
 - EC2
 - S3 (limited)

Now add users:

- Rahul
- Priya

✅ Both automatically get the same access

♦ Why Use IAM Groups

- ⚡ Easier management (no need to assign permissions individually)
-  Consistency (all users in a role have the same access)
-  Scalable (just add/remove users from group)

Key Points

- Groups contain only users (not roles or other groups)
- Permissions are assigned via policies
- A user can belong to multiple groups

- ❖ **Policy** - An IAM policy is a JSON document that specifies who can access which resources and what actions they can perform.

Basic Structure of a Policy

A policy has these main parts:

- **Version** → Policy language version
- **Statement** → One or more permission rules
 - **Effect** → Allow or Deny
 - **Action** → What actions are permitted (e.g., `s3:GetObject`)
 - **Resource** → Which resources (e.g., specific S3 bucket)
 - **Condition** (*optional*) → Extra restrictions

Types of IAM Policies

Managed Policies

- Predefined or reusable
- Two types:
 - AWS-managed (e.g., `AmazonS3ReadOnlyAccess`)
 - Customer-managed (created by you)

Inline Policies

- Attached directly to:
 - A user
 - A group
 - A role
- Not reusable.